

2. PRELIMINARIES

2.1. Divisors and line bundles. Let X be an irreducible complex variety of dimension n . The group of k -**cycles** on X is

$$Z_k(X) = \{\mathbb{Z}\text{-linear combinations of subvarieties of dimension } k\}.$$

We will often denote $Z_k(X)$ by $Z^{n-k}(X)$. The group of **Weil divisors** is just

$$\mathrm{WDiv}(X) = Z_{n-1}(X).$$

The group of **Cartier divisors** is

$$\mathrm{Div}(X) := \Gamma(X, \mathcal{K}_X^* / \mathcal{O}_X^*).$$

Here $\mathcal{K}_X^*$ denotes the sheaf of non-zero rational functions on X and $\mathcal{O}_X^*$ is the sheaf of nonvanishing regular functions. Therefore, a Cartier divisor D can be given by specifying an open cover U_i of X and sections $f_i \in \Gamma(U_i, \mathcal{K}_X^*(U_i))$ such that $f_i/f_j \in \Gamma(U_i \cap U_j, \mathcal{O}_X^*(U_i \cap U_j))$. Given divisors $D, D' \in \mathrm{Div}(X)$, then by definition $D + D' = \{U_i, f_i f'_i\}$. There is a natural homomorphism of groups

$$\mathrm{Div}(X) \longrightarrow \mathrm{WDiv}(X)$$

given by $D \longrightarrow [D] = \sum \mathrm{ord}_V(D) \cdot [V]$ where $\mathrm{ord}_V(D)$ is the order of vanishing of the f_i along the codimension 1 subvariety. This homomorphism is injective when X is normal and an isomorphism when X is smooth.

Exercise 2.1. Let $X = \mathrm{Spec} k[x, y, z]/(xy - z^2)$ be the quadric cone given by the equation $xy - z^2$. Show that $\mathrm{WDiv}(X) \neq \mathrm{Div}(X)$. (In fact $\mathrm{WDiv}(X) = \mathbb{Z}/2\mathbb{Z}$ and $\mathrm{Div}(X) = 0$.)

There is a short exact sequence

$$0 \longrightarrow \mathcal{O}_X^* \longrightarrow \mathcal{K}_X^* \longrightarrow \mathcal{K}_X^* / \mathcal{O}_X^* \longrightarrow 0.$$

Therefore, there is a natural homomorphism

$$\mathrm{Div}(X) \cong H^0(X, \mathcal{K}_X^* / \mathcal{O}_X^*) \longrightarrow H^1(X, \mathcal{O}_X^*) \cong \mathrm{Pic}(X)$$

which sends a divisor $D = \{U_i, f_i\}$ to the line bundle $\mathcal{O}_X(D) = \{U_i \cap U_j, f_i/f_j\}$. The kernel of this map is the group of **principal divisors**

$$\mathrm{Princ}(X) \subset \mathrm{Div}(X).$$

Two divisors D, D' are linearly equivalent $D \sim D'$ if $D - D' \in \mathrm{Princ}(X)$. We denote by

$$|D| := \{D' \mid D' \sim D\}$$

the **complete linear series** of D . Notice that $D = \mathbb{P}H^0(X, \mathcal{O}_X(D))$. Given any subvector space $V \subset H^0(X, \mathcal{O}_X(D))$ there is a rational map

$$\phi_V : X \dashrightarrow \mathbb{P}V = |V|$$

given by

$$x \longrightarrow [s_0(x) : s_1(x) : \dots : s_n(x)]$$

where $s_0, \dots, s_n$ is a basis of V . This map is only defined away from the base locus of V and it sends x to the hyperplane of divisors containing x .

We say that D is **very ample** if ϕ_D is an embedding. D is **ample** if there is a positive integer m such that mD is ample.

2.2. Rules of coherent cohomology. Let X be a variety (over an algebraically closed field k), and $\mathcal{F}$ a quasi-coherent sheaf¹, then there exist k -vector spaces $H^i(X, \mathcal{F})$ such that:

- (1) For any homomorphism of $\mathcal{O}_X$ -modules $a : \mathcal{F} \longrightarrow \mathcal{G}$, there is an induced k -linear map

$$a_* : H^i(X, \mathcal{F}) \longrightarrow H^i(X, \mathcal{G});$$

- (2) $H^0(X, \mathcal{F}) \cong \Gamma(X, \mathcal{F})$;

- (3) If $0 \longrightarrow \mathcal{F}' \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}'' \longrightarrow 0$ is a short exact sequence, of quasi-coherent sheaves on X , then there is a coboundary map $d : H^i(X, \mathcal{F}'') \longrightarrow H^{i+1}(X, \mathcal{F}')$ such that the sequence

$$\dots H^i(X, \mathcal{F}') \longrightarrow H^i(X, \mathcal{F}) \longrightarrow H^i(X, \mathcal{F}'') \longrightarrow H^{i+1}(X, \mathcal{F}') \dots$$

is exact;

- (4) $H^i(X, \mathcal{F}) = 0$ for all $i > \dim X$;
- (5) If $\mathcal{F}$ is coherent², X is proper then $H^i(X, \mathcal{F})$ is finite dimensional and we denote its dimension by $h^i(X, \mathcal{F})$.
- (6) $h^i(\mathbb{P}^r, \mathcal{O}_{\mathbb{P}^r}(n)) = 0$ for all $n \in \mathbb{Z}$ and all $1 \leq i \leq r - 1$;
 $h^0(\mathbb{P}^r, \mathcal{O}_{\mathbb{P}^r}(n)) = \binom{r+n}{n}$ if $n \geq 0$ and 0 if $n < 0$; $h^r(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^r}(-n - r - 1)) = h^0(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^r}(n))$. (cf. [Hartshorne] III.5.1.

Recall that the **Euler characteristic** of a coherent sheaf $\mathcal{F}$ is

$$\chi(X, \mathcal{F}) = \sum_{i \geq 0} h^i(X, \mathcal{F}).$$

¹i.e. it is locally isomorphic to the cokernel of a homomorphism of locally free sheaves

²i.e. it is locally isomorphic to the cokernel of a homomorphism of locally free sheaves of finite rank

Theorem 2.2 (Asymptotic Riemann-Roch). *Let X be an irreducible projective variety of dimension n , and $D \in \text{Div}(X)$. Then $\chi(X, \mathcal{F} \otimes \mathcal{O}_X(mD))$ is a polynomial of degree at most n in m such that*

$$\chi(X, \mathcal{F} \otimes \mathcal{O}_X(mD)) = \text{rank}(\mathcal{F}) \frac{D^n}{n} m^n + O(m^{n-1}).$$

Proof. See Hirzebruch-Riemann-Roch:

$$\chi(X, \mathcal{F} \otimes \mathcal{O}_X(mD)) = \text{ch}(\mathcal{F} \otimes \mathcal{O}_X(mD)) \cdot Td(X).$$

□

2.3. Serre's Theorems. Recall that a coherent sheaf $\mathcal{F}$ is globally generated if the homomorphism

$$H^0(X, \mathcal{F}) \otimes \mathcal{O}_X \longrightarrow \mathcal{F}$$

is surjective.

Theorem 2.3 (Serre). *Let X be a projective scheme and D be a very ample line bundle and $\mathcal{F}$ a coherent sheaf. Then there is an integer $n_0 > 0$ such that for all $n \geq n_0$, $\mathcal{F} \otimes \mathcal{O}_X(nD)$ is globally generated.*

Proof. [Hartshorne] II §5. The idea is that we may assume that $X = \mathbb{P}^N$. Since $\mathcal{F}$ is coherent, it is locally generated by finitely many sections of $\mathcal{O}$. Each local section is the restriction of some global section of $\mathcal{O}_{\mathbb{P}^N}(n)$ for $n \gg 0$. By compactness, we only need finitely many such sections. □

Theorem 2.4 (Serre Vanishing). *Let X be a projective scheme and D be a very ample line bundle and $\mathcal{F}$ a coherent sheaf. Then there is an integer $n_0 > 0$ such that for all $n \geq n_0$ and all $i > 0$,*

$$H^i(X, \mathcal{F} \otimes \mathcal{O}_X(nD)) = 0.$$

Proof. (cf. [Hartshorne] III.5.2) We may assume that $X = \mathbb{P}^N$ (replace $\mathcal{F}$ by $\phi_{D*}\mathcal{F}$). The Theorem is clear if $\mathcal{F}$ is a finite direct sum of sheaves of the form $\mathcal{O}_{\mathbb{P}^N}(q)$. We can find a short exact sequence of coherent sheaves

$$0 \rightarrow \mathcal{K} \rightarrow \oplus \mathcal{O}_{\mathbb{P}^N}(q_i) \rightarrow \mathcal{F} \rightarrow 0$$

(eg. use (2.3)). We now consider the exact sequence

$$\cdots \oplus H^i(\mathcal{O}_{\mathbb{P}^N}(q_i + n)) \rightarrow H^i(\mathcal{F} \otimes \mathcal{O}_{\mathbb{P}^N}(n)) \rightarrow H^{i+1}(\mathcal{K} \otimes \mathcal{O}_{\mathbb{P}^N}(n)) \cdots$$

and proceed by descending induction on i so that we may assume that $h^{i+1}(\mathcal{K} \otimes \mathcal{O}_{\mathbb{P}^N}(n)) = 0$ for $n \gg 0$ and $i \geq 0$. Since $h^i(\mathcal{O}_{\mathbb{P}^N}(q_i + n)) = 0$ for $n \gg 0$ and $i \geq 0$, we have $h^i(\mathcal{F} \otimes \mathcal{O}_{\mathbb{P}^N}(n)) = 0$ as required for $n \gg 0$ and $i \geq 0$. □

Recall that we also have the following

Proposition 2.5. *The following are equivalent*

- (1) D is ample;
- (2) mD is ample for all $m > 0$;
- (3) mD is ample for some $m > 0$;
- (4) for any coherent sheaf $\mathcal{F}$, there exists an integer $m > 0$ such that $\mathcal{F} \otimes \mathcal{O}_X(nD)$ is globally generated.

Proof. Exercise. (See [Hartshorne] II §7). □

2.4. The cone of effective 1-cycles. For any $D \in \text{Div}(X)$ and $C = \sum n_i C_i \in Z_1(X)$ we define

$$D \cdot C = \sum n_i (D \cdot C_i)$$

by considering the normalization $\nu_i : \tilde{C}_i \rightarrow C_i$ and letting

$$D \cdot C_i = \deg(\nu_i^* \mathcal{O}_X(D)).$$

Two Cartier divisors D, D' are **numerically equivalent** $D \equiv D'$ if

$$D \cdot C = D' \cdot C \quad \forall C \text{ irreducible curves on } X.$$

Two curves C, C' are **numerically equivalent** $C \equiv C'$ if

$$C \cdot D = C' \cdot D \quad \forall D \in \text{Div}(X).$$

Remark 2.6. Recall that there is a short exact sequence

$$0 \rightarrow \mathbb{Z} \rightarrow \mathcal{O}_X \rightarrow \mathcal{O}_X^* \rightarrow 0$$

inducing a short exact sequence

$$\text{Pic}^0(X) \rightarrow \text{Pic}(X) \rightarrow \text{NS}(X) \rightarrow 0.$$

here $\text{Pic}^0(X) \cong H^1(X, \mathcal{O}_X)/H^1(X, \mathbb{Z})$ is the group of topologically trivial line bundles and the **Neron Severi group** is

$$\text{NS}(X) \cong \text{Imm}(H^1(X, \mathcal{O}_X^*) \rightarrow H^2(X, \mathbb{Z})).$$

Since the group $\text{Div}(X)/\equiv$ is a quotient of $\text{NS}(X)$, it is a free abelian group of finite rank which we denote by $\rho(X)$ the **Picard number** of X .

We will adopt the following notation:

$$N_1(X) = (Z_1(X)/\equiv) \otimes \mathbb{R},$$

$$N^1(X) = (Z^1(X)/\equiv) \otimes \mathbb{R},$$

so that $\rho(X) = \dim_{\mathbb{R}} N_1(X) = \dim_{\mathbb{R}} N^1(X)$. We define the **cone of effective 1-cycles** to be

$$NE(X) \subset N_1(X)$$

to be the cone generated by $\{\sum n_i C_i \text{ s. t. } n_i \geq 0\}$. Given a morphism $f : X \rightarrow Y$ and an irreducible curve $C \subset X$, we let $f_*(C) = df(C)$ where $d = \deg(C \rightarrow f(C))$. If $f(C)$ is a point, then we set $f_*C = 0$. One sees that

$$f^*D \cdot C = D \cdot f_*C \quad \forall D \in \text{Div}(Y).$$

Extending by linearity, we get an injective homomorphism

$$f^* : N^1(Y) \rightarrow N^1(X)$$

and surjective homomorphisms

$$f_* : N_1(X) \rightarrow N_1(Y), \quad NE(X) \rightarrow NE(Y).$$

Let W be a subvariety of X of codimension i , and $D \in \text{Div}(X)$ then we may also define the intersection numbrs $D^i \cdot W \in \mathbb{Z}$. Suppose that this has been done for all $j > i$, then pick a very ample divisor H such that $D + H$ is also very ample. Then $(D + H)^i \cdot W$ is just the degree of the image of W via $\phi_{H^0(X, \mathcal{O}_X(D+H))}$ and $D^j H^{i-j} W$ have been previously defined (as the codimension of $H^{i-j} W$ is $j > i$). Hence $D^i \cdot W$ is also defined. We have the following important results:

Theorem 2.7. *[Nakai-Moishezon criterion] Let D be a Cartier divisor on a projective scheme X , then D is ample if and only if for any $0 \leq i \leq n - 1$ and any subvariety W of codimension i , one has $D^i \cdot W > 0$.*

It is enough to assume that X is proper.

Proof. (cf. [KM98] 1.37) We may assume that X is irreducible. Clearly, if D is ample, then $D^i \cdot W > 0$. For the converse implication, we proceed by induction on $n = \dim X$. When $\dim X = 1$, the Theorem is obvious. So we may assume that $D|_Z$ is ample for all proper closed subschemes $Z \subsetneq X$.

Claim 1. $h^0(X, \mathcal{O}_X(D)) > 0$ for some $k > 0$ (actually $\kappa(D) = n$).

We choose a very ample divisor B such that $D+B$ is very ample. Let $A \in |D+B|$ be a general member. Consider the short exact sequence

$$0 \rightarrow \mathcal{O}_X(kD - B) \rightarrow \mathcal{O}_X(kD) \rightarrow \mathcal{O}_B(kD) \rightarrow 0.$$

Since $\mathcal{O}_B(kD)$ is ample, for all $k \gg 0$ we have that $h^i(B, \mathcal{O}_B(kD)) = 0$ for all $i > 0$, and so

$$h^i(\mathcal{O}_X(kD - B)) = h^i(\mathcal{O}_X(kD)) \quad \text{for } i \geq 2 \text{ and } k \gg 0.$$

The same argument applied to the short exact sequence

$$0 \rightarrow \mathcal{O}_X(kD - B) \rightarrow \mathcal{O}_X((k+1)D) \rightarrow \mathcal{O}_A((k+1)D) \rightarrow 0$$

shows that

$$h^i(\mathcal{O}_X(kD - B)) = h^i(\mathcal{O}_X((k+1)D)) \quad \text{for } i \geq 2 \text{ and } k \gg 0.$$

Putting this together, we see that for $i \geq 2$ and $k \gg 0$, one has $h^i(\mathcal{O}_X(kD)) = h^i(\mathcal{O}_X((k+1)D))$ and so the number $h^i(\mathcal{O}_X(kD))$ is constant. But then for $k \gg 0$

$$\begin{aligned} h^0(X, \mathcal{O}_X(kD)) &\geq h^0(X, \mathcal{O}_X(kD)) - h^1(X, \mathcal{O}_X(kD)) \\ &= \chi(X, \mathcal{O}_X(kD)) + (\text{constant}) = D^n/n \cdot k^n + O(k-1). \end{aligned}$$

Claim 2. $\mathcal{O}_X(kD)$ is generated by global sections for some $k > 0$.

Fix a non-zero section $s \in H^0(X, \mathcal{O}_X(mD))$. Let S be the divisor defined by s . Consider the short exact sequence

$$0 \longrightarrow \mathcal{O}_X((k-1)mD) \longrightarrow \mathcal{O}_X(kmD) \longrightarrow \mathcal{O}_S(kmD) \longrightarrow 0.$$

By induction $\mathcal{O}_S(kmD)$ is generated by global sections and so it suffices to show that for $k \gg 0$, the homomorphism $H^0(X, \mathcal{O}_X(kmD)) \longrightarrow H^0(S, \mathcal{O}_S(kmD))$ is surjective. Arguing as in Claim 1, one sees that $h^1(X, \mathcal{O}_X(kmD))$ is a decreasing sequence and is hence eventually constant as required.

Conclusion of the proof. $\phi_{kD} : X \longrightarrow \mathbb{P}^N$ is a finite morphism for $k \gg 0$. If in fact C is a curve contracted by ϕ_{kD} , then $kD \cdot C = \mathcal{O}_{\mathbb{P}^N}(1) \cdot \phi_{kD}(C) = 0$ which contradicts $C \cdot D > 0$. The Theorem now follows as $kD = \phi_{kD}^* \mathcal{O}_{\mathbb{P}^N}(1)$ is ample (as it is the pullback of an ample line bundle via a finite map). $\square$

We have the following useful corollary:

Theorem 2.8. [Kleiman's Theorem] *Let X be a proper variety, D a nef divisor. Then $D^{\dim Z} \cdot Z \geq 0$ for all irreducible subvarieties $Z \subset X$.*

Proof. (cf. [Lazarsfeld05] 1.4.9) We assume that X is projective (Chow's Lemma) and irreducible. When $\dim X = 1$, the Theorem is clear. By induction on $n = \dim X$, we may assume that

$$D^{\dim Z} \cdot Z \geq 0 \quad \forall Z \subset X \text{ irreducible of } \dim Z < n,$$

and we must show that $D^n \geq 0$. Fix H an ample divisor and consider the polynomial

$$P(t) := (D + tH)^n \in \mathbb{Q}[t].$$

We must show that $P(0) \geq 0$. For $1 \leq k \leq n$, the coefficient of t^k is

$$D^{n-k} H^k \geq 0.$$

Assume that $P(0) < 0$, then one sees that $P(t)$ has a unique real root $t_0 > 0$.

For any rational number $t > t_0$, one sees that

$$(D + tH)^{\dim Z} \cdot Z > 0 \quad \forall Z \subset X \text{ irreducible of } \dim Z \leq n,$$

and so by (2.7), $D + tH$ is ample. We write

$$P(t) = Q(t) + R(t) = D \cdot (D + tH)^{n-1} + tH \cdot (D + tH)^{n-1}.$$

As $D + tH$ is ample for $t > t_0$, one has that $(D + tH)^{n-1}$ is an effective 1-cycle, so $Q(t) > 0$ for all rational numbers $t > t_0$ and so $Q(t_0) \geq 0$. One sees that all the coefficients of $R(t)$ are non-negative and the coefficient of t^n is $H^n > 0$. It follows that $R(t_0) > 0$ and so $P(t_0) > 0$ which is the required contradiction. $\square$

Exercise 2.9. *Let X be a projective variety, H an ample divisor on X . A divisor D on X is nef if and only if $D + \epsilon H$ is ample for all rational numbers $\epsilon > 0$.*

Exercise 2.10. *Let X be a projective variety, H an ample divisor on X . A divisor D on X is ample if and only if there exists an $\epsilon > 0$ such that*

$$\frac{D \cdot C}{H \cdot C} \geq \epsilon$$

for all irreducible curves $C \subset X$.

Theorem 2.11. *[Nakai's Criterion] Let D be a divisor on a projective scheme X . D is ample if and only if $D \cdot Z > 0$ for any $Z \in \overline{NE}(X) - \{0\}$.*

Proof. Exercise. $\square$

Theorem 2.12. *[Seshadri's Criterion] Let D be a divisor on a projective scheme X . D is ample if and only if there exists an $\epsilon > 0$ such that*

$$C \cdot D \geq \epsilon \operatorname{mult}_x(C)$$

for all $x \in C \subset X$.

Proof. Assume that D is ample, then there exists an integer $n > 0$ such that nD is very ample, but then

$$nD \cdot C \geq \operatorname{mult}_x(C)$$

for all $x \in C \subset X$.

For the reverse implication, we proceed by induction. Therefore, we may assume that for any irreducible subvariety $Z \subsetneq X$, the divisor $D|_Z$ is ample and so $D^{\dim Z} \cdot Z > 0$. By (2.7), it is enough to show that $D^n > 0$. Let

$$\mu : X' \longrightarrow X$$

be the blow up of X at a smooth point. Then $\mu^*D - \epsilon E$ is nef. In fact, for any curve $C' \subset X'$ we either have $C = \mu(C')$ is a curve and then

$$(\mu^*D - \epsilon E) \cdot C' = D \cdot C - \epsilon \operatorname{mult}_x C \geq 0,$$

or $\mu(C') = x$ and then since $C' \subset E \cong \mathbb{P}^{n-1}$ and $\mathcal{O}_E(E) = \mathcal{O}_{\mathbb{P}^{n-1}}(-1)$

$$(\mu^*D - \epsilon E) \cdot C' = \epsilon \deg C' > 0.$$

But then, by (2.8)

$$(\mu^*D - \epsilon E)^n = D^n - \epsilon^n \geq 0$$

and this completes the proof. □